

UNIVERSITY OF CALIFORNIA SAN DIEGO

Generative AI for Music and Audio

A dissertation submitted in partial satisfaction of the
requirements for the degree Doctor of Philosophy

in

Computer Science

by

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Committee in charge:

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2024

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University of California San Diego

2024

DEDICATION

*To my wife Eltha, my parents Nancy and David, and my pooh bear;
without whom nothing would have been possible.*

EPIGRAPH

Without music, life would be a mistake.

—Friedrich Nietzsche

TABLE OF CONTENTS

Dissertation Approval Page	iii
Dedication	iv
Epigraph	v
Table of Contents	vi
List of Figures	ix
List of Tables	xii
Acknowledgements	xiii
Vita	xvii
Abstract of the Dissertation	xix
Chapter 1 Introduction	1
1.1 Multitrack Music Generation	3
1.2 Assistive Music Creation Tools	4
1.3 Multimodal Learning for Audio and Music	5
1.4 Dissertation Organization	6
Chapter 2 MusPy: A Toolkit for Symbolic Music Generation	7
Abstract	7
2.1 Introduction	7
2.2 Related Work	9
2.3 MusPy	10
2.3.1 MusPy Music class and I/O interfaces	11
2.3.2 Dataset management	12
2.3.3 Representations	12
2.3.4 Model evaluation tools	14
2.3.5 Summary	15
2.4 Dataset Analysis	15
2.5 Experiments and Results	17
2.5.1 Experiment settings	18
2.5.2 Autoregressive models on different datasets	18
2.5.3 Cross-dataset generalizability	18
2.5.4 Effects of combining heterogeneous datasets	20
2.6 Conclusion	22
Chapter 3 Multitrack Music Transformer	23
Abstract	23
3.1 Introduction	23
3.2 Related Work	25

3.3	Proposed Method	26
3.3.1	Data Representation	26
3.3.2	Model	28
3.4	Results	30
3.4.1	Experiment Setup	30
3.4.2	Subjective Listening Test	30
3.4.3	Objective Evaluation	32
3.4.4	Musical Self-attention	32
3.5	Conclusion	35
Chapter 4 Towards Automatic Instrumentation by Learning to Separate Parts in Symbolic Multitrack Music		36
	Abstract	36
4.1	Introduction	37
4.2	Prior Work	39
4.3	Problem Formulation	40
4.4	Models	41
4.5	Baseline Models	42
4.5.1	Zone-based algorithm	42
4.5.2	Closest-pitch algorithm	42
4.5.3	Multilayer perceptron (MLP)	43
4.6	Data	43
4.7	Experiments	44
4.7.1	Implementation details	44
4.7.2	Qualitative results and error analysis	45
4.7.3	Quantitative results	45
4.7.4	Effect of input features	47
4.7.5	Effects of time encoding	49
4.7.6	Effects of data augmentation	49
4.8	Discussion	50
4.9	Conclusion	51
Chapter 5 Deep Performer: Score-to-Audio Music Performance Synthesis		52
	Abstract	52
5.1	Introduction	52
5.2	Methods	54
5.2.1	Alignment model	55
5.2.2	Synthesis model	55
5.2.3	Inversion model	57
5.3	Data	57
5.4	Experiments & Results	58
5.4.1	Implementation details	58
5.4.2	Qualitative and quantitative results	59
5.4.3	Subjective listening test	60
5.4.4	Ablation study	62
5.5	Conclusion	62
	Appendices	64

Chapter 6	CLIPSep: Learning Text-queried Sound Separation with Noisy Unlabeled Videos	68
	Abstract	68
6.1	Introduction	69
6.2	Related Work	71
6.3	Method	73
6.3.1	CLIPSep—Learning text-queried sound separation without labeled data	73
6.3.2	Noise invariant training—Handling noisy data in the wild	75
6.4	Experiments	77
6.4.1	Experiments on clean data	77
6.4.2	Experiments on noisy data	79
6.4.3	Examining the effects of the noise regularization level γ	81
6.5	Discussions	83
6.6	Conclusion	85
6.7	Acknowledgements	85
	Appendices	86
Chapter 7	CLIPSonic: Text-to-Audio Synthesis with Unlabeled Videos and Pretrained Language-Vision Models	98
	Abstract	98
7.1	Introduction	99
7.2	CLIPSonic	101
7.3	Experimental setup	103
7.4	Results	105
7.4.1	Objective Evaluation Results	105
7.4.2	Subjective Listening Test Results	108
7.5	Conclusion	109
	Appendices	110
Chapter 8	Conclusion	113
8.1	Future Directions	114
8.2	Broader Impacts	116

LIST OF FIGURES

Figure 1.1.	Looking back to the past, how music and technology interacts has always been a two-way process.	2
Figure 1.2.	An overview of the three main directions of my research.	3
Figure 2.1.	An example of a learning-based music generation system.	8
Figure 2.2.	System diagram of MusPy.	10
Figure 2.3.	Examples of (a) training data preparation and (b) result writing pipelines using MusPy.	11
Figure 2.4.	Two internal processing modes for iterating over a MusPy Dataset object.	13
Figure 2.5.	Length distributions for different datasets.	16
Figure 2.6.	Initial-tempo distributions for different datasets (those without tempo information are not presented).	16
Figure 2.7.	Key distributions for different datasets.	17
Figure 2.8.	Log-perplexities for different models on different datasets, sorted by the values for the LSTM model.	19
Figure 2.9.	Log-perplexities for the LSTM model versus dataset size in hours.	19
Figure 2.10.	Cross-dataset generalizability results.	21
Figure 3.1.	An example of the proposed representation.	27
Figure 3.2.	Illustration of the proposed MMT model.	29
Figure 3.3.	Mean relative attention gains (a) $\tilde{\gamma}_k^{beat}$, (b) $\tilde{\gamma}_k^{position}$ and (c) $\tilde{\gamma}_k^{pitch}$ (see Section 3.4.4 for definitions) of a trained MMT model.	34
Figure 4.1.	Proposed pipeline.	38
Figure 4.2.	Example of the Bach chorales dataset— <i>Wer nur den lieben Gott läßt walten</i> , BWV 434, measures 1–5.	45
Figure 4.3.	Hard excerpt in the string quartets dataset—Beethoven’s <i>String Quartet No. 11 in F minor, Op. 95</i> , movement 1, measures 72–83.	46
Figure 4.4.	Hard excerpt in the game music dataset— <i>Theme of Universe</i> from <i>Miracle Ropit’s Adventure in 2100</i>	46

Figure 4.5.	Hard excerpt in the pop music dataset— <i>Blame It On the Boogie</i> by The Jacksons.	47
Figure 4.6.	<i>Quando Quando Quando</i> by Tony Renis—(a) original instrumentation and the versions produced by (b) the online LSTM model without entry hints and (b) the offline BiLSTM model with entry hints.	50
Figure 5.1.	An overview of the proposed three-stage pipeline for score-to-audio music performance synthesis.	53
Figure 5.2.	An illustration of the proposed synthesis model.	56
Figure 5.3.	An example of the constant-Q spectrogram of the first 20 seconds of a violin recording and the estimated onsets (white dots) and durations (green lines).	58
Figure 5.4.	Examples of the alignments predicted by (a) the constant-tempo baseline model and (b) Deep Performer, our proposed model. (c) shows the input score.	59
Figure 5.5.	Examples of the mel spectrograms, in log scale, synthesized by our proposed model for (a) violin and (c) piano. (b) and (d) show the input scores for (a) and (c), respectively.	60
Figure 5.6.	Examples of the mel spectrograms, in log scale, synthesized by (a) the baseline model, (b) our proposed synthesis model, and (d) our proposed synthesis model without the note-wise positional encoding. (c) and (e) show the waveforms for (b) and (d), respectively. (f) shows the input score.	61
Figure 6.1.	An illustration of modality transfer.	70
Figure 6.2.	An illustration of the proposed CLIPSep model for $n = 2$	74
Figure 6.3.	An illustration of the proposed CLIPSep-NIT model for $n = 2$	75
Figure 6.4.	Mean SDR and standard errors of the models trained and tested on different modalities.	78
Figure 6.5.	Example results of the proposed CLIPSep-NIT model with $\gamma = 0.25$ on the MUSIC ⁺ dataset.	82
Figure 6.6.	Effects of the noise regularization level γ for the proposed CLIPSep-NIT model—mean SDR for the (a) MUSIC ⁺ and (b) VGGSound-Clean ⁺ evaluations, and (c) the total mean noise head activation, $\sum_{i=1}^n \text{mean}(\hat{M}_i^N)$, on the validation set.	83
Figure 7.1.	We learn the text-audio correspondence by leveraging the audio-visual correspondences in videos and the multimodal representation learned by pretrained language-vision models.	100

Figure 7.2.	Proposed CLIP Sonic model.	102
Figure 7.3.	Objective evaluation results on VGGSound and MUSIC.....	106
Figure 8.1.	An overview of my future research directions.	115

PREVIEW

LIST OF TABLES

Table 2.1.	Comparisons of MIDI, MusicXML and the proposed MusPy formats.....	12
Table 2.2.	Comparisons of datasets currently supported by MusPy.	13
Table 2.3.	Comparisons of representations supported by MusPy.....	14
Table 3.1.	Comparisons of related transformer-based music models.....	25
Table 3.2.	Performance comparison of our proposed model against the baseline models.	31
Table 3.3.	Objective evaluation results.....	32
Table 4.1.	Statistics of the four datasets considered in this paper.	43
Table 4.2.	Comparison of our proposed models and baseline algorithms.	48
Table 4.3.	Effects of input features to the online LSTM model.	48
Table 4.4.	Comparisons of time encoding and data augmentation strategies for the online LSTM model.	49
Table 5.1.	Comparisons of the final MSE between the synthesized mel spectrograms and the ground truths, in log scales.	60
Table 5.2.	Results of the subjective listening test.	61
Table 6.1.	Comparisons of related work in sound separation.	72
Table 6.2.	Results on the MUSIC dataset.....	77
Table 6.3.	Results of the MUSIC ⁺ and VGGSound-Clean ⁺ evaluations (see Section 6.4.2).	79
Table 7.1.	Evaluation results on VGGSound and MUSIC datasets, evaluated at $w = 1.5$	107
Table 7.2.	Cosine similarities between various query embeddings.....	107
Table 7.3.	Listening test results for text-to-audio synthesis (MOS).	108
Table 7.4.	Listening test results for image-to-audio synthesis (MOS).	108

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- *Chapter 3, in full, is a reprint of the material as it appears in “Multitrack Music Transformer” by Hao-Wen Dong, Ke Chen, Shlomo Dubnov, Julian McAuley and Taylor Berg-Kirkpatrick, which was published in the Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) in 2023. The dissertation author was the primary investigator and author of this paper.*
- *Chapter 4, in full, is a reprint of the material as it appears in “Towards Automatic Instrumentation by Learning to Separate Parts in Symbolic Multitrack Music” by Hao-Wen Dong, Chris Donahue, Taylor Berg-Kirkpatrick and Julian McAuley, which was published in the Proceedings of the International Society for Music Information Retrieval Conference (ISMIR) in 2021. The dissertation author was the primary investigator and author of this paper.*
- *Chapter 5, in full, is a reprint of the material as it appears in “Deep Performer: Score-to-Audio Music Performance Synthesis” by Hao-Wen Dong, Cong Zhou, Taylor Berg-Kirkpatrick and Julian McAuley, which was published in the Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) in 2022. The dissertation author was the primary investigator and author of this paper.*
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Hao-Wen Dong, Xiaoyu Liu, Jordi Pons, Gautam Bhattacharya, Santiago Pascual, Joan Serrà, Taylor Berg-Kirkpatrick, and Julian McAuley, “CLIPsonic: Text-to-Audio Synthesis with Unlabeled Videos and Pretrained Language-Vision Models,” *IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)*, 2023.

Hao-Wen Dong, Gunnar A. Sigurdsson, Chenyang Tao, Jiun-Yu Kao, Yu-Hsiang Lin, Anjali Narayan-Chen, Arpit Gupta, Tagyoung Chung, Jing Huang, Nanyun Peng, and Wenbo Zhao, “CLIPSynth: Learning Text-to-audio Synthesis from Videos using CLIP and Diffusion Models,” *CVPR Workshop on Sight and Sound (WSS)*, 2023.

Hao-Wen Dong, Ke Chen, Shlomo Dubnov, Julian McAuley, and Taylor Berg-Kirkpatrick, “Multitrack Music Transformer,” *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023.

Hao-Wen Dong, Naoya Takahashi, Yuki Mitsufuji, Julian McAuley, and Taylor Berg-Kirkpatrick, “CLIPSep: Learning Text-queried Sound Separation with Noisy Unlabeled Videos,” *International Conference on Learning Representations (ICLR)*, 2023.

Ke Chen, Hao-Wen Dong, Yi Luo, Julian McAuley, Taylor Berg-Kirkpatrick, Miller Puckette, and Shlomo Dubnov, “Improving Choral Music Separation through Expressive Synthesized Data from Sampled Instruments,” *International Society for Music Information Retrieval Conference (ISMIR)*, 2022.

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Hao-Wen Dong, Chris Donahue, Taylor Berg-Kirkpatrick, and Julian McAuley, “Towards Automatic Instrumentation by Learning to Separate Parts in Symbolic Multitrack Music,” *International Society for Music Information Retrieval Conference (ISMIR)*, 2021.

Sachinda Edirisooriya, Hao-Wen Dong, Julian McAuley, and Taylor Berg-Kirkpatrick, “An Empirical Evaluation of End-to-End Polyphonic Optical Music Recognition,” *International Society for Music Information Retrieval Conference (ISMIR)*, 2021.

Yin-Cheng Yeh, Wen-Yi Hsiao, Satoru Fukayama, Tetsuro Kitahara, Benjamin Genschel, Hao-Min Liu, Hao-Wen Dong, Yian Chen, Terence Leong, and Yi-Hsuan Yang, “Automatic Melody Harmonization with Triad Chords: A Comparative Study,” *Journal of New Music Research (JNMR)*, 50(1):37–51, 2021.

Hao-Wen Dong, Ke Chen, Julian McAuley, and Taylor Berg-Kirkpatrick, “MusPy: A Toolkit for Symbolic Music Generation,” *International Society for Music Information Retrieval Conference (ISMIR)*, 2020.

Hao-Wen Dong and Yi-Hsuan Yang, “Convolutional Generative Adversarial Networks with Binary Neurons for Polyphonic Music Generation,” *International Society for Music Information Retrieval Conference (ISMIR)*, 2018.

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ABSTRACT OF THE DISSERTATION

Generative AI for Music and Audio

by

Hao-Wen Dong

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Generative AI has been transforming the way we interact with technology and consume content. In the next decade, AI technology will reshape how we create audio content in various media, including music, theater, films, games, podcasts, and short videos. In this dissertation, I introduce the three main directions of my research centered around *generative AI for music and audio*: 1) multitrack music generation, 2) assistive music creation tools, and 3) multimodal learning for audio and music. Through my research, I aim to answer the following two fundamental questions: 1) *How can AI help professionals or amateurs create music and audio content?* 2) *Can AI learn to create music in a way similar to how humans learn music?* My long-term goal is to lower the barrier of entry for music composition and democratize audio content creation.

Chapter 1

Introduction

*AI is the study of how to make computers do things
at which, at the moment, people are better.*

—Elaine Rich and Kevin Knight

Generative AI has been transforming the way we interact with technology and consume content. The recent success of large language model-based chatbots (e.g., OpenAI's ChatGPT¹ and Google's Gemini²), AI assistants (e.g., GitHub and Microsoft Copilot) and text-to-image generation systems (e.g., Adobe Firefly,³ Midjourney⁴ and Stable Diffusion⁵) showcases how AI-powered technology can be integrated into professional workflows and boost human productivity. In the next decade, generative AI technology will also reshape how we create audio content in the \$2.3 trillion global entertainment industry, including the music, film, TV, podcast and gaming sectors. Take AI-powered music creation for example: On one hand, we have witnessed major progress in automatic music composition (Briot et al., 2017; Huang et al., 2020), which has long been considered as a grand challenge of AI. On the other hand, our expectations of *AI Music* today has expanded to cover the whole music creation process—from composition, arrangement, sound production, recording to mixing (De Man et al., 2019). With a growing momentum in both academia and industry, AI-powered audio creation has been gaining attention in the broader AI community.

¹<https://chat.openai.com/>

²<https://gemini.google.com/>

³<https://firefly.adobe.com/>

⁴<https://www.midjourney.com/>

⁵<https://stability.ai/stable-image>

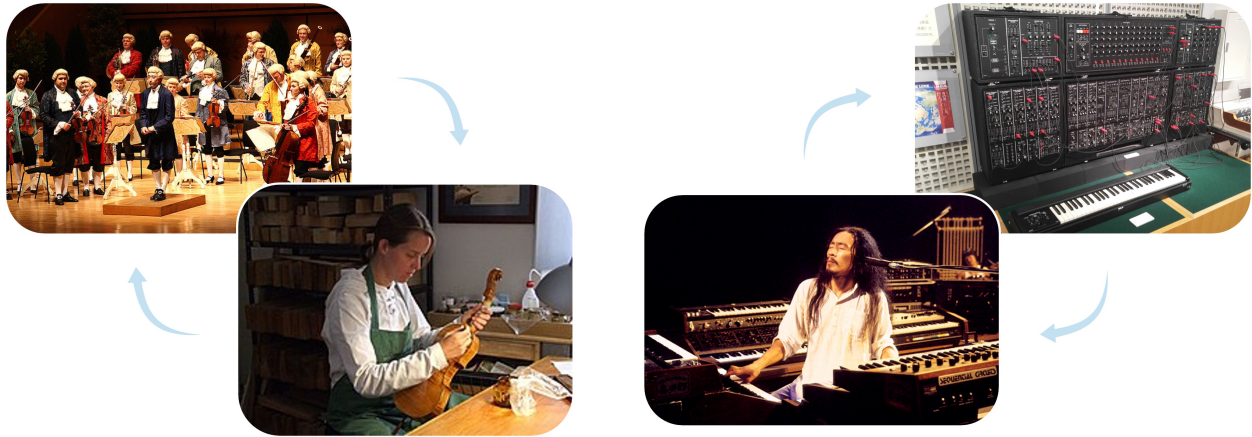


Figure 1.1. Looking back to the past, how music and technology interacts has always been a two-way process. (Left) the violin-making industry grows with the classical music, and together create the golden age of classical music. (Right) the invention and development of synthesizers and drum machines helped popularize electronic music. Image sources (from left to right): 1) Mozart83, Public domain, via Wikimedia Commons, 2) Hildegard Dodel, Public domain, via Wikimedia Commons, 3) yan, CC BY-SA 4.0, via Wikimedia Commons, and 4) taken at Hamamatsu Museum of Musical Instruments, August 2019.

My research springs from two fundamental questions: 1) *How can AI help professionals or amateurs create music and audio content?* 2) *Can AI learn to create music in a way similar to how humans learn music?* From a musical perspective, technology has always been a driving factor of music evolution. For example, the study of acoustics and musical instrument making fostered the development of classical music; the invention of synthesizers and drum machines helped popularize electronic music. I am thus interested in exploring how the latest AI technology can empower artists to create novel contents. From a technical perspective, music possesses a unique complexity in that music follows rules and patterns while being creative and expressive at the same time. I am thus fascinated about the idea of building intelligent systems that can learn, create and play music like humans do. I envision the future development of AI Music to be a two-way process—*new technology creates new music; new music inspires new technology.*

Motivated by this belief, I study a wide range of topics centered around *Generative AI for Music and Audio*, including multitrack music generation (Dong et al., 2018a; Dong et al., 2017; Dong and Yang, 2018; Dong et al., 2023a; Xu et al., 2023; Liu et al., 2018a), automatic instrumentation (Dong et al., 2021), automatic arrangement (Dong et al., 2018a; Liu et al., 2018a), automatic harmonization (Yeh et al., 2021), music performance synthesis (Dong et al.,

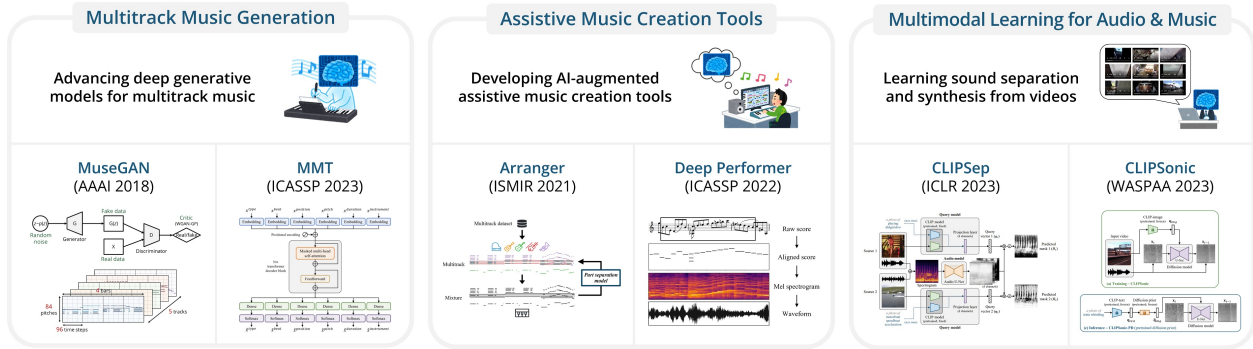


Figure 1.2. An overview of the three main directions of my research.

2022), text-queried sound separation (Dong et al., 2023d), text-to-audio synthesis (Dong et al., 2023c; Dong et al., 2023b) and symbolic music processing software (Dong et al., 2018b; Dong et al., 2020).

My research can be categorized into three main directions, as shown in Figure 1.2: 1) **multitrack music generation**—*advancing deep generative models for multitrack music*, 2) **assistive music creation tools**—*developing AI tools that can help musicians and amateurs create music*, and 3) **multimodal learning for audio and music**—*learning sound separation and synthesis from noisy videos*.

1.1 Multitrack Music Generation

Researchers have been working on automatic music composition for decades, and it has long been viewed as a grand challenge of AI. I started this thread of research on multitrack music generation in 2017. Back then, prior work on deep learning-based music generation had focused on generating melodies, lead sheets (i.e., melodies and chords) or four-part chorales Briot et al., 2017. However, modern pop music often consists of multiple instruments or tracks. To make deep learning technology applicable in modern music production workflow, it is important to modernize deep learning models for multitrack music generation.

Witnessing the lack of infrastructure for symbolic music generation when conducting these research projects, I developed libraries for processing symbolic music to consolidate the infrastructure of music generation research (Dong et al., 2018b; Dong et al., 2020). The Python toolkits I developed to process symbolic music for machine learning applications

have been widely used in the field. With these libraries, researchers can easily download commonly used datasets programmatically and be liberated from reimplementing tedious data processing routines. The toolkits also allowed me to conduct *the first large-scale experiment that measures the cross-dataset generalizability of deep neural networks for music* (Dong et al., 2020). This work will be presented in Chapter 2.

Moreover, I study efficient music generation model with an eye to enabling real time improvisation or near real time creative applications. In Dong et al., 2023a, I proposed the Multitrack Music Transformer (MMT) model that achieves comparable performance with state-of-the-art systems, landing in between two recently proposed models in a subjective listening test, while achieving substantial speedups and memory reductions over both. Further, I presented *the first systematic analysis of musical self-attention*, where I showed that the trained model learns a relative attention in certain aspects of music. With the great success brought by large-scale generative pretraining in natural language processing, my ongoing work aims to scale this method up using the largest ever symbolic music dataset containing more than one million scores, which I collected and compiled from the MuseScore forum. This work will be presented in Chapter 3.

1.2 Assistive Music Creation Tools

Music creation today is still largely limited to professional musicians for it requires a certain level of knowledge in music theory, music notation and music production tools. Apart from generating new music content from scratch, another line of my research focuses on developing AI-augmented tools to assist amateurs to create and perform music. My long-term goal along this research direction is to lower the entry barrier of music composition and make music creation accessible to everyone.

For example, in (Dong et al., 2021), I developed *the first deep learning model for automatic instrumentation*. Instrumentation refers to the process where a musician arranges a solo piece for a certain ensemble such as a string quartet or a rock band. This can be challenging for amateur composers as it requires domain knowledge of each target instrument. In this work, I proposed a new machine learning model that can produce convincing instrumentation for a solo piece by framing this problem as a sequential multi-class classification

problem. Such an automatic instrumentation system can suggest potential instrumentation for amateur composers, especially useful when arranging for an unfamiliar ensemble. Further, the proposed model can empower a musician to play multiple instruments on a single keyboard at the same time. This work will be presented in Chapter 4.

Another example is my work on music performance synthesis (Dong et al., 2022). While synthesizers play a critical role and are intensively used in modern music production, existing synthesizers either requires an input with expressive timing or allows only monophonic inputs. In light of the similarities between text-to-speech (TTS) and score-to-audio synthesis, I showed in this work that we can adapt a state-of-the-art TTS model for music performance synthesis. Moreover, I proposed a novel mechanism to enable polyphonic music synthesis. This work represents *the first deep learning based polyphonic synthesizer that can synthesize a score into a natural, expressive performance*. This work will be presented in Chapter 5.

1.3 Multimodal Learning for Audio and Music

The third line of my research focuses on multimodal learning for audio and music. Sound is an integral part of movies, dramas, documentaries, podcasts, games, short videos and audiobooks. In these media, audio and music production tools need to interact with inputs from other modalities such as text and images, and thus multimodal models are critical in enabling controllable creation tools for music and audio in these applications.

Along this direction, I have worked on text-queried sound separation (Dong et al., 2023d) and text-to-audio synthesis (Dong et al., 2023c; Dong et al., 2023b). Unlike existing work that relies on a large amount of paired audio-text data, I explore a new direction of approaching bimodal learning for text and audio through leveraging the visual modality as a bridge. The key idea behind my study is to combine the naturally-occurring audio-visual correspondence in videos and the multimodal representation learned by contrastive language-vision pretraining (CLIP). Based on this idea, I developed *the first text-queried sound separation model that can be trained without any text-audio pairs* (Dong et al., 2023d). Text-queried sound separation aims to separate a specific sound out from a mixture of sounds given a text query, which has many downstream applications in audio post-production